



Department of Computer Science

CSI481 Database Systems

Assignment 2

Please Note: This task should be carried out in the same groups which were for Assignment 1.

Due: Friday 23 October 2026 by 17:00 hrs

Part 2: Logical Design, Database and Application Programs

The purpose of the exercise is to design and build a relational database together with some application programs that interact with the database. It hopefully will complement the lectures by providing practical experience.

Your Task:

Using the conceptual model (model solution given on Moodle) from the first part of the assignment do the following:

1. Logical Design and Database Implementation

Translate the UML design into a logical schema for implementation in a Relational DBMS such as MySQL or ORACLE (or any other available DBMS that you find suitable for such an application). You will need to consider the storage parameters offered by that DBMS to create a suitable database for this application.

Implement the database by creating the relations and constraints and adding enough test data into the relations.

2. Application programs

Use Java to produce application programs that interact with the database through a graphical user interface to perform the following tasks:

1. Add a new GP, vaccine, patient and parent.
2. Register a vaccination with a vaccine, patient and GP that are already registered in the database (with no need to enter their details again).
3. Schedule the next booster vaccination for a patient (assuming they took the first dose) and appoint with a GP for the scheduled vaccination.

4. Give a list of a patients and their vaccination details including vaccine name, booster number (if any), date of vaccination and GP details of the GP who carried out the vaccination.
5. Produce reports as stated at the end of the scenario below.

Requirements

This scenario was adapted from

Paul Beynon-Davies (2004). Database Systems. Palgrave Macmillan

A system is required to record details of infant immunisation in a health region. Every infant in the region is required to have a course of general vaccinations against diseases such as whooping cough and diptheria. Patients are identified by a unique birth certificate number and details such as the name, date of birth and Omang number of the parent or guardian of the infant are also recorded.

Each vaccination is for a single vaccination type such as mumps, rubella, etc., and is given to a single patient. However, every infant is given a number of booster injections of certain vaccination types at periodic intervals. The date of each vaccination is recorded.

Vaccinations are given by General Practitioners (Gps). A general practice usually has many GPs. Each GP works for only one general practice and usually has many patients on his or her list. Each infant is on the list of only one doctor. Practices are identified by practice names and GPs are identified by unique GP numbers. The name of each doctor and the number of vaccination patients on each doctor's list also needs to be recorded.

Besides general update operations, the following processes must be supported by the system:

- The production of appointment letters
- A work-list by General Practice for a given week
- An audit list indicating the number of vaccinations by type conducted during the previous six-month period

Submission

Submit two files: a Java executable file and the logical model (logical.pdf) (1 submission per group) in Moodle by the due date, with the database username/password hard-coded to allow running it from the lecturer's side without having to log on.